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● *Original Contribution*

CONSERVATIVE TREATMENT OF UVEAL MELANOMA: LOCAL RECURRENCE AFTER PROTON BEAM THERAPY

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Twenty-three of 1006 (2.3%) uveal melanoma patients treated with proton beam therapy at the Harvard Cyclotron Laboratory between July 1975 and December 31, 1986 received additional treatment for documented (15 patients) or suspected (eight patients) tumor growth in the irradiated eye. Growth within the initially irradiated volume was documented at Massachusetts Eye and Ear Infirmary in 12 patients. Documented growth occurred in nine of 665 (1.4%) patients with small and intermediate size tumors, at times after treatment ranging from 6 to 48 months (median 16 months), and in three of 341 (.9%) patients with large tumors at 7, 11, and 12 months after treatment. Melanoma growing totally outside the treated volume was also documented in three additional patients at 7, 9, and 45 months; two of these were thought to be "ring melanomas". Eight patients had the treated eye removed elsewhere for suspected tumor growth. The additional treatment in these 23 patients was conservative in nine patients (repeat proton irradiation in five and laser photocoagulation in four). Thirteen underwent immediate enucleation and one had orbital exenteration. Ultimately, 17 of the 23 eyes (74%) were removed. Estimated probability of local control of the melanoma within the irradiated eye at 60 months was 96.3 ± 1.5%. Dose distributions to the 12 patients with documented local failure within the irradiated volume were analyzed. Ten tumors recurred marginally in an area receiving less than the prescribed dose of 70 CGE (CGE = Cobalt Gray Equivalents = proton Gy × RBE 1.1), whereas only two recurred in the volume receiving full dose. Based on these data, it appears that a dose of 70 CGE in five fractions is associated with very high rates of local control in human uveal melanoma. It is reasonable to consider initiating studies using a lower total dose or a more protracted course, to determine if some of the observed complications are dose-related.

Ocular melanoma, Proton beam therapy.

INTRODUCTION

There has been ample recent documentation that vision is being preserved in most conservatively treated uveal melanoma patients (5, 6, 9, 15, 22, 36, 37, 40). Survival does not appear to have been compromised by allowing the treated eye to remain in place, rather than resorting initially to radical treatment, that is, enucleation at the time of diagnosis or documentation of growth (1, 27, 29, 39). Some eyes, however, have had to be removed after treatment, primarily because of complications in patients treated conservatively with external beam charged particle therapy (4, 6, 11, 22, 34, 37, 41) or radioisotope plaque therapy (2, 15, 16, 27, 30, 31, 32, 35, 36). Conservative treatment with radiation therapy has been successful in achieving local control of the irradiated tumor in the majority of cases (4, 5, 6, 15, 22, 27, 31, 32, 33, 35).

This study was undertaken to study local control rates for uveal melanomas treated with the proton beam, by determining how many patients received additional treat-

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